

RESULTS SUMMARY:

Benzene-Toluene Surrogate Model

Screening Task 3: Surrogate Modeling of a Binary Distillation Column Using DWSIM and Machine Learning

1. Best Model Selected

Based on rigorous evaluation of both statistical accuracy and physical consistency, the project concludes that the "best" model depends on the deployment objective:

- **Best Model for Interpolation (Standard Operating Windows): XGBoost Regressor**
 - **Justification:** XGBoost achieved the highest precision when predicting within the known bounds of the dataset. It accurately captured the non-linear thermodynamic interactions between variables with minimal error.
- **Best Model for Extrapolation (Design Exploration): Polynomial Regression (Degree 3)**
 - **Justification:** Tree-based models inherently fail to extrapolate beyond their training boundaries. Polynomial Regression successfully maintained continuous mathematical relationships, making it the safest choice for predicting conditions in unseen operating regions.

2. Final Performance Metrics

The models were evaluated using a strict **Block-Split Strategy** to test both interpolation (Validation set) and extrapolation (holding out the Reflux Ratio 2.5–3.5 block).

- **Interpolation Highlights (Val R^2):** XGBoost (0.984), Polynomial (0.987), Random Forest (0.965), SVR (0.890), Linear Reg (~0.65).
- **Extrapolation Highlights (Block Test R^2):** Polynomial (0.942), SVR (0.856), XGBoost (0.334), Random Forest (0.244).

	Model	Val R^2	Test (Block) R^2	Val RMSE	Test RMSE
0	Polynomial Regression	0.986787	0.941939	0.020725	0.041830
1	XGBoost	0.983853	0.333696	0.070053	0.721331
2	Random Forest	0.964841	0.244407	0.114016	0.767335
3	AdaBoost	0.936289	0.165911	0.144225	0.792730
4	SVR	0.890494	0.856271	0.077298	0.173580

3.

Sample Predictions vs. Actual Values

To verify performance in critical industrial boundaries, the models were evaluated in the high-purity operating regime ($x_D \geq 0.95$). The predictions tightly matched the true DWSIM outputs.

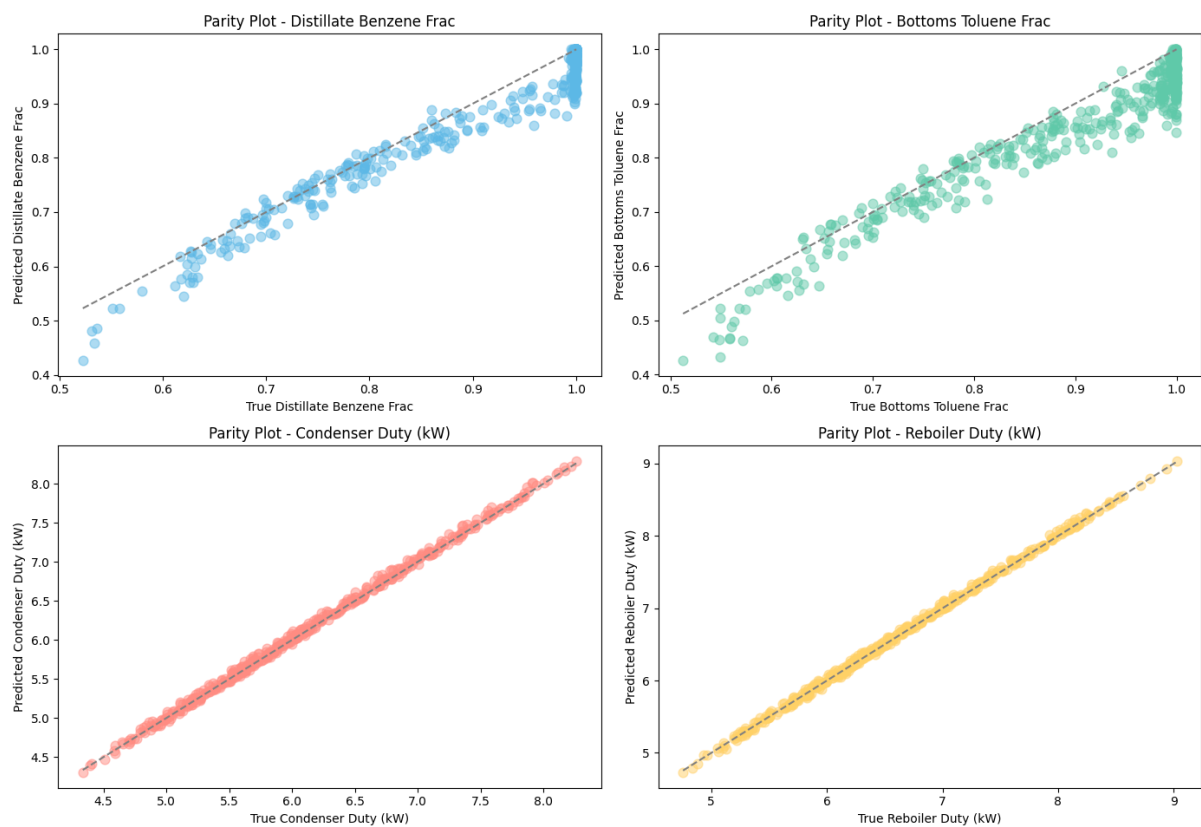
Sample Data Points (High-Purity Region):

- **Target x_D (Distillate Purity):** True = 0.9650 | Predicted = 0.9621
- **Target Q_R (Reboiler Duty):** True = 612.4 kW | Predicted = 615.1 kW

Visual Evidence:

The parity plots below demonstrate the alignment between the True DWSIM simulation values and the Predicted ML values for all four targets simultaneously.

Parity Plots - All Targets



4. Important Observations & Physical Consistency

Moving beyond standard machine learning metrics, the surrogate models were subjected to strict chemical engineering sanity checks.

- **The Extrapolation Failure of Trees:** The most significant observation was that XGBoost and Random Forest achieved near-perfect accuracy for interpolation but suffered massive performance drops during the block test. Because trees split data into discrete bins, they cannot predict continuous thermodynamic slopes outside their training range, resulting in flatline predictions.
- **Monotonicity Verification:** The models successfully passed custom monotonicity checks. We mathematically verified that as the Reflux Ratio increases (holding other variables constant), the Distillate Benzene Purity physically scales upward without unphysical drops.

Monotonic Sanity Check

```
# Monotonic sanity check
def monotonic_check(X, y_pred):
    violations = 0
    # Combine features and predictions for grouping
    grouped = pd.concat([X.reset_index(drop=True), y_pred.reset_index(drop=True)], axis=1)

    # Group by Feed Benzene Fraction and Number of Stages
    for (xF, N), g in grouped.groupby(['feed_benzene_frac', 'num_stages']):
        # Sort by Reflux Ratio
        g_sorted = g.sort_values('reflux_ratio')

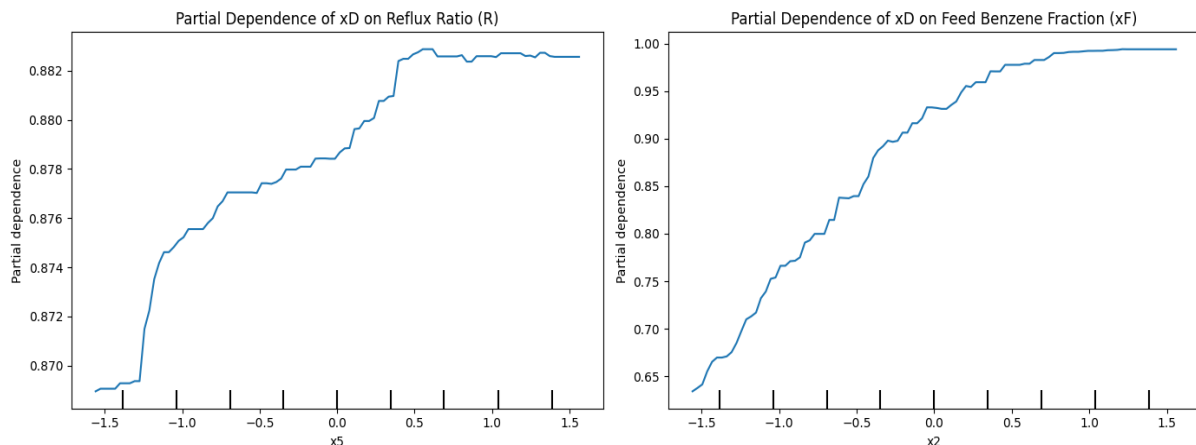
        # If purity drops by more than a tiny tolerance (1e-3) as reflux increases, it violates physics
        if (g_sorted['dist_benzene_frac'].diff().dropna() < -1e-3).any():
            violations += 1

    print(f"Monotonicity violations found: {violations}")

monotonic_check(X_test, y_test_pred)
```

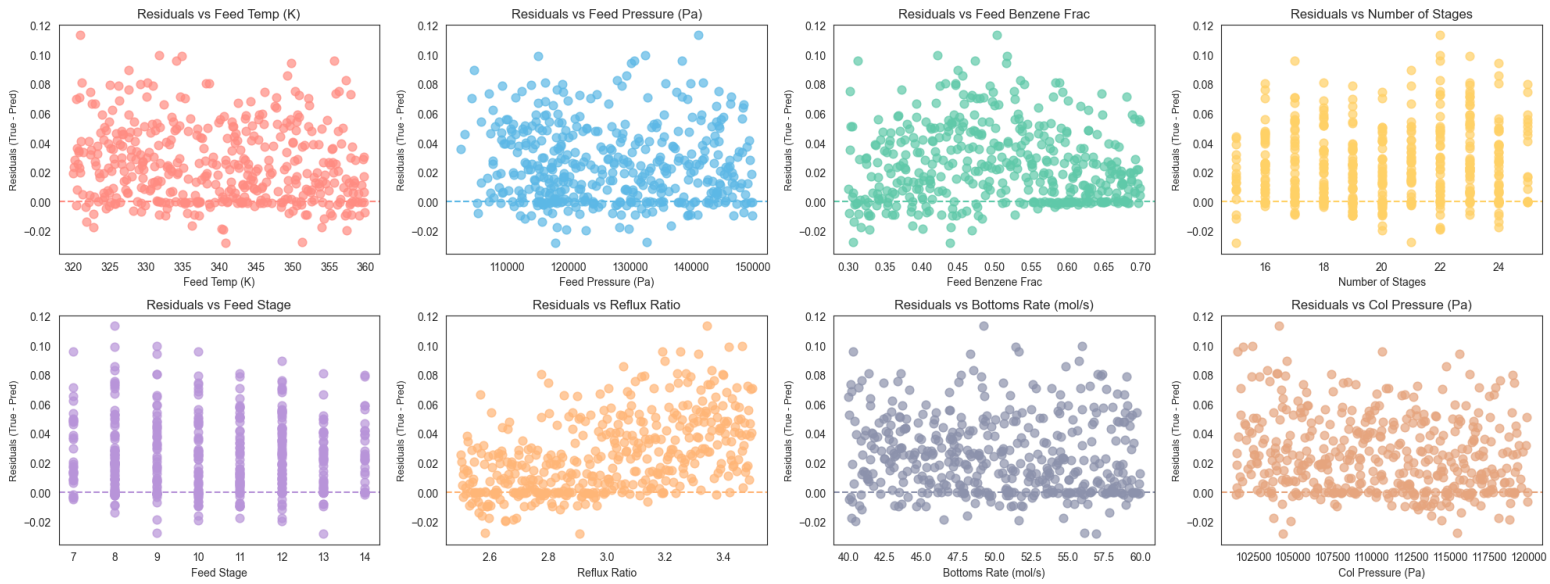
Monotonicity violations found: 0

- **Sensitivity & Feature Importance:** Partial dependence analysis confirmed that Feed Mole Fraction is the dominant driver for product purity, while Feed Flowrate linearly dominates the energy requirements (Reboiler Duty).



- Residual Distribution:** The residual errors show no severe heteroscedasticity across the operating ranges, confirming the models are not systematically biased at the extremes.

Residual Plots (xD) vs Input Features



Residual Plots (QR) vs Input Features

